

BASE BERNARDO O'HIGGINS, Antarctica

WMO: 890590

Lat: 63.321S

Lon: 57.899W

Elev: 10

StdP: 101.20

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -18.9	(c) -17.2	(d) -22.0	(e) 0.5	(f) -18.0	(g) -20.1	(h) 0.6	(i) -16.1	(j) 41.1	(k) -13.4	(l) 32.4	(m) -11.9	(n) 16.0	(o) 150	(p) 1.081

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 2.5	(c) 3.6	(d) 1.7	(e) 2.9	(f) 1.5	(g) 2.4	(h) 1.2	(i) 2.4	(j) 2.9	(k) 2.0	(l) 2.4	(m) 1.6	(n) 2.1	(o) 6.3	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
2.0	4.4	2.4	1.5	4.2	2.1	1.1	4.1	1.7	14.1	3.1	13.3	2.5	12.5	2.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.3	23.8	20.3	-21.9	7.3	3.0	2.9	-24.1	9.4	-25.8	11.1	-27.5	12.7	-29.7	14.8
			-21.7	4.1	2.9	1.2	-23.8	5.0	-25.5	5.7	-27.1	6.4	-29.2	7.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.3	0.9	0.4	-1.1	-3.0	-4.3	-6.6	-7.9	-7.0	-5.7	-3.8	-1.8	0.0		
	DBStd	4.29	1.08	1.43	2.22	3.16	3.33	4.14	4.63	4.18	4.32	3.09	1.97	1.36		
	HDD10.0	4870	283	269	343	391	443	499	554	526	471	427	353	311		
	HDD18.3	7912	542	502	601	641	702	749	812	784	721	686	603	569		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	6.7	5.4	5.4	6.5	6.9	6.6	7.8	7.3	7.5	7.4	7.3	6.5	6.2		
(15) Precipitation	PrecAvg	919	69	73	96	82	76	62	65	68	80	97	83	65		
	PrecMax	1250	107	114	136	150	120	103	147	128	111	132	128	107		
	PrecMin	725	45	53	60	46	40	27	22	18	35	63	41	29		
	PrecStd	121	15	15	18	22	18	17	25	27	19	16	22	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	4.3	3.4	2.2	1.9	0.9	0.7	0.6	1.2	1.2	3.0	4.2		
		MCWB	2.9	2.4	1.7	1.3	0.6	0.0	-0.3	-0.1	0.2	0.1	1.4	2.0		
	2%	DB	3.6	3.1	2.5	1.6	0.9	0.0	-0.1	-0.1	0.3	0.6	1.6	3.0		
		MCWB	1.7	1.9	1.4	0.8	0.3	-0.5	-0.7	-0.6	-0.4	-0.1	0.4	1.2		
	5%	DB	2.9	2.6	2.0	1.0	0.3	-0.6	-0.7	-0.6	-0.1	0.2	1.0	2.2		
		MCWB	1.4	1.6	1.2	0.4	-0.3	-1.1	-1.2	-1.1	-0.7	-0.4	0.1	0.8		
	10%	DB	2.4	2.1	1.5	0.6	-0.2	-1.1	-1.4	-1.1	-0.6	-0.2	0.5	1.6		
		MCWB	1.1	1.3	0.8	0.0	-0.7	-1.7	-1.9	-1.6	-1.1	-0.8	-0.3	0.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.1	2.6	1.8	1.2	0.2	0.1	0.1	0.4	0.6	1.7	2.5		
		MCDB	4.2	3.5	2.8	2.0	1.5	0.4	0.3	0.3	0.8	0.8	2.2	3.6		
	2%	WB	2.4	2.4	1.9	1.1	0.5	-0.4	-0.5	-0.4	-0.2	0.1	0.9	1.7		
		MCDB	3.0	2.7	2.2	1.3	0.7	-0.1	-0.2	-0.2	0.1	0.4	1.3	2.4		
	5%	WB	1.8	1.9	1.5	0.6	-0.1	-0.9	-1.0	-0.9	-0.5	-0.3	0.4	1.1		
		MCDB	2.4	2.3	1.8	0.9	0.2	-0.6	-0.7	-0.7	-0.2	0.0	0.8	1.8		
	10%	WB	1.4	1.5	1.0	0.1	-0.6	-1.6	-1.8	-1.5	-0.9	-0.6	-0.1	0.7		
		MCDB	2.0	1.9	1.4	0.4	-0.3	-1.2	-1.4	-1.2	-0.6	-0.3	0.4	1.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	2.5	2.4	2.9	3.2	3.8	4.3	4.8	4.8	4.1	3.3	2.7	2.5		
		MCDBR	3.5	3.1	3.1	3.1	3.4	4.0	5.1	4.9	3.1	3.0	3.0	3.3		
	5% WB	MCWBR	2.7	2.5	2.7	3.0	3.2	3.9	4.8	5.0	3.0	2.9	2.7	2.7		
		MCWBR	3.0	2.7	2.7	2.8	3.1	3.5	4.2	4.7	3.0	2.8	2.8	3.0		
(40) Clear-Sky Solar Irradiance	taub		0.293	0.290	0.275	0.249	0.184	0.157	0.172	0.241	0.266	0.288	0.288	0.293		
	taud		2.383	2.398	2.410	2.331	2.029	1.919	1.987	2.205	2.293	2.306	2.340	2.351		
	Ebn at Noon		977	929	852	695	450	254	447	675	840	918	977	990		
	Edh at Noon		108	94	74	50	27	13	26	54	81	102	112	115		
(44) All-Sky Solar Radiation	RadAvg		5.54	3.90	2.13	0.91	0.26	0.08	0.17	0.70	1.94	3.73	5.49	6.27		
	RadStd		0.35	0.22	0.16	0.06	0.02	0.00	0.01	0.05	0.12	0.20	0.31	0.44		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page